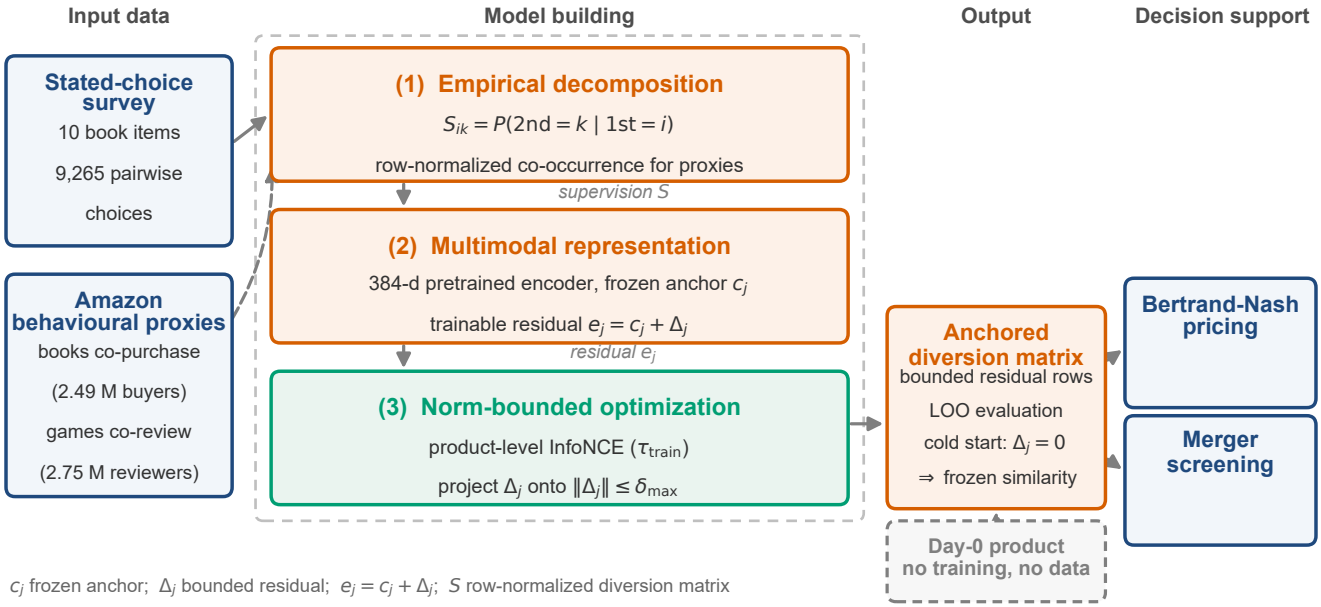

Input data
Model building
Output
Decision support
Stated-choice survey
10 book items
9,265 pairwise
choices
Amazon behavioural proxies
books co-purchase
(2.49 M buyers)
games co-review
(2.75 M reviewers)
(1) Empirical decomposition
$S_{ik} = P(2\text{nd} = k \mid 1\text{st} = i)$
row-normalized co-occurrence for proxies
supervision S
(2) Multimodal representation
384-d pretrained encoder, frozen anchor c_j
trainable residual $e_j = c_j + \Delta_j$
residual e_j
(3) Norm-bounded optimization
product-level InfoNCE (τ_{train})
project Δ_j onto $\|\Delta_j\| \le \delta_{\max}$
Anchored diversion matrix
bounded residual rows
LOO evaluation
cold start: $\Delta_j = 0$
⇒ frozen similarity
Day-0 product
no training, no data
Bertrand-Nash pricing
Merger screening
c_j frozen anchor; Δ_j bounded residual; $e_j = c_j + \Delta_j$; S row-normalized diversion matrix